

150,000nits Full-color Native RGB Single Junction 3,386PPI Micro-OLED

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Abstract

In this study, we independently developed a knowledge system named uNEEDXR™, successfully realizing a full-color micro-OLED device with a brightness of 150,000 nits on a silicon-based backplane. This knowledge system was developed over nine years and innovatively know-how and integrates expertise across design, processes, manufacturing, equipment, and material, overcoming the performance limitations of traditional micro-OLED architectures. The system enables high brightness, high pixel density, low power consumption, high contrast ratio, high color saturation, and tunable energy distribution (including viewing angle, wavelength, and bandwidth). Additionally, it meets customer requirements for reliability and lifespan. This technology provides a scalable production solution for near-eye display applications in augmented reality.

Author Keywords

High brightness; OLED; Micro-OLED; Native RGB

1. Introduction

Augmented Reality (AR) applications require full-color, high brightness, high contrast, and low power micro-displays to be adaptable across various environments. In AR applications, projected images must be visible in diverse lighting conditions, necessitating high brightness to ensure that data and images remain clear even in ambient light or when using low efficiency optical systems. Additionally, high contrast is essential to prevent glare in the display area and minimize interference from environments.

The key considerations for head mounted displays (HMD) include making low power consumption, lightweight design, and a compact architecture for user acceptance.

Micro OLEDs utilize single-crystal silicon (Si) wafers as the substrate. Compared to traditional glass-based OLEDs, Si-based backplanes offer higher carrier mobility, enabling to achieve smaller pixel sizes. The manufacturing process of Micro OLEDs involves depositing the OLED emission layers onto the single crystal silicon substrate, forming a fully integrated display module.

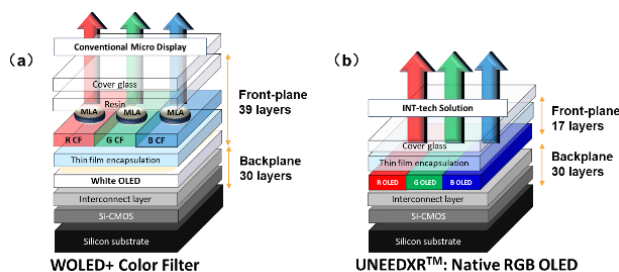


Figure 1. Structure of (a) conventional white OLED(WOLED) with color filter(CF) in comparison with a (b)

uNEEDXR™ (Native RGB OLED) with RGB emitter layers.

uNEEDXR™: ultra-high pixel density for Near-EyE Display-XR

Conventional AMOLED micro-displays typically employ a white OLED layer covering the entire display area, combined with a color filter (CF) positioned above the thin film encapsulation, as illustrated in Fig 1 (a). The white OLED emits light through the CF to achieve red, green, and blue (RGB) colors. However, CFs absorb more than 70% of the emitted light, significantly reducing optical efficiency. As a result, current commercially available displays using this approach have a brightness limitation of approximately 10,000 cd/m², which remains insufficient for AR applications [1]-[4].

In recent years, direct patterned full color AMOLED technology has emerged in the market. Some approaches employ ultra fine patterned masks on silicon wafers to directly fabricate RGB OLED pixels [5], while others use dry and wet etching techniques to achieve patterned full color OLED displays, as illustrated in Fig 1 (b). The former method faces challenges such as mask clogging and the difficulty of aligning ultra fine masks, while the latter involves complex and multi-processes. Additionally, during the fabrication process, it is necessary to address potential damage caused by moisture and oxygen in wet processes, as well as plasma in dry etching. These challenges can lead to efficiency degradation, increased driving voltage, and a significant reduction in device lifespan, issues that are still being actively addressed.

In this presentation, we introduce our proprietary uNEEDXR™ technology, which enables high brightness (150,000 cd/m²) and high resolution (3,386 PPI) 0.39" native RGB device. Through innovative advancements in process steps and environmental control, our proprietary patterning process has demonstrated excellent performance.

2. Methods

uNEEDXR™ devices were fabricated on a 0.39" XGA resolution (1024 x 768) CMOS backplane (0.18 micron) featuring a 7.5 μm pitch (3386 PPI). Each 7.5 μm pixel is comprised of three RGB sub pixels with a diameter length of 3.5 μm and is patterned with a highly reflective anode. The anode gap between the sub pixels is 0.6 μm (with an aperture ratio of 51%), as illustrated in Fig. 3. The native RGB process revealed here, common hole injection layers and hole transport layers are deposited over the emission area using an open metal mask [6]. Following this process, the red, green or blue OLED layers are directly patterned individually for each color on top of the anode electrode.

All OLED organic materials used in this study were commercially available and easily accessible. After individually patterning and depositing the red, green, and blue emissive layers along with their respective electron transport layers, then the cathode was deposited over the entire emission area using an open metal shadow mask. Following the OLED deposition process, thin-film

encapsulation was then applied to complete the device fabrication. uNEEDXR™ native RGB process significantly simplifies the traditional method of using white light combined with color filters. Additionally, uNEEDXR™ native RGB technique was with high brightness, high color saturation, and low power consumption.

Emission characteristics (luminance, color coordinates and spectra) were measured by a Spectrophotometer CS2000. Microscope images of 0.39" display were taken using a Keyence VHX-7000. In this work, we compare the performance of our native RGB 0.39" device with that of the global leading company.

The process begins with externally sourced 8-inch wafers, where the anode metal layer is initially deposited. The anode is then patterned and etched to define the desired electrode structure. Subsequently, a Pixel Definition Layer (PDL) is deposited over the anode and similarly patterned to create openings for OLED deposition.

The uNEEDXR™ technology (which requires a strictly controlled oxygen- and moisture-free environment) is employed to selectively expose the first color pixel regions, followed by evaporation and deposition of the first color OLED material. This uNEEDXR™ process (maintained under the same controlled atmosphere) is then repeated to sequentially open the second and third color pixel regions, with deposition of their corresponding OLED layers.

After the RGB OLED deposition is completed, the cathode layer is evaporated across the entire surface. A thin-film encapsulation (TFE) layer is then deposited to protect the OLED structure from moisture and oxygen. The device is further encapsulated by laminating a cover glass under vacuum to ensure a hermetic seal.

The processed wafer is then diced into individual chips, after which the bonding pads are exposed to enable electrical connections. A flexible printed circuit (FPC) is bonded to the pads, and the final devices undergo light-on testing to verify functionality and performance.

This streamlined approach leverages uNEEDXR™ technology for precise, sequential RGB patterning, enabling efficient OLED deposition while maintaining high resolution and yield, as illustrated in Fig. 2.

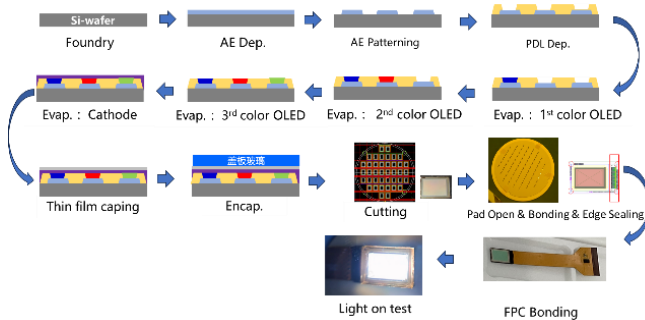


Figure 2. uNEEDXR™ Process flow

3. Results and Discussion

The product recently unveiled by INT-Tech is a full-color 0.39" XGA display, with following specifications, as shown in Table 1 :

Table 1. Product specification

Screen size	0.39" Full-color
Resolution	XGA (1024*RGB*768)
Pixel density	3386PPI (pixel pitch: 7.5um)
Maximum brightness	150,000nits±5% (OLED biased @9V)
Power consumption	<3000mW @150,000 nits ±5%
Contrast ratio	>100,000:1 @150,000 nits ±5%
Color gamut	DCI-P3: 95%
Brightness uniformity	>90% (nine points average)

In near-eye display experiences, high pixel density displays are required to eliminate the screen door effect and deliver more realistic imaging [7][8]. In the specification table (Table 1), the 0.39-inch Micro-OLED has a pixel density of 3386 PPI (PPI refers to the number of pixels per inch), a pixel pitch of 7.5 μm , a subpixel diameter of 3.3 μm , and an aperture ratio of 58%. By using uNEEDXR™'s patterning technology, the RGB subpixels can achieve high pixel density without color mixing (as shown in Figure 3).

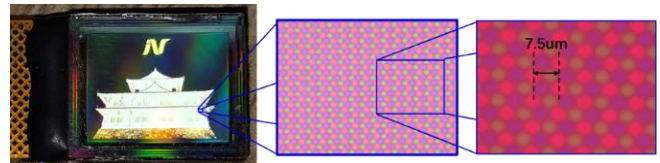


Figure 3. Pixel Density 3386PPI (pixel pitch: 7.5um)

We used the CS2000 luminance meter to measure the brightness, chromaticity coordinates, and spectrum of the 0.39-inch full-color product. The maximum full-color brightness measured is 151,605 nits, with chromaticity coordinates (CIEx, CIEy) = (0.376, 0.352). The peak wavelength of red is 625 nm, green is 525 nm, and blue is 463 nm (as shown in Figure 4).

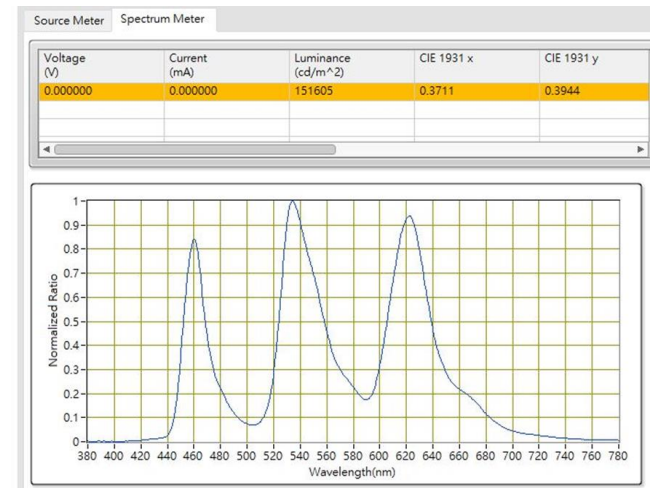


Figure 4. Full-color 0.39"XGA measured by CS2000

For the 0.39-inch XGA product, the maximum brightness at L255 grayscale can exceed 150,000 nits, while the minimum brightness

at L0 grayscale is less than 0.0001 nits. Therefore, the contrast ratio of L255/L0 can reach over 100,000,000:1, as shown in Figure 5 (Native RGB has no color filter and appears like a mirror in black scenes).

The benefit of high contrast is that the display remains clear through AR glasses under both daytime and nighttime backgrounds [9].



Figure 5. Contrast ratio test result

In terms of power consumption, we compared it with commercially available 0.39-inch XGA full-color silicon-based OLEDs using white light and color filters. At the same brightness of 3,000 nits, the power consumption of native RGB is only one-fifth of that of WOLED + CF. At the same power consumption of 550 mW, the brightness of native RGB is 11 times higher

than that of WOLED + CF. Therefore, the power efficiency at high brightness and the same power level is more than 11 times greater (as shown in Figure 6).

The advantage of low power consumption allows for longer battery life and reduces battery size, enabling a smaller and lighter AR glasses form factor. This leads to a better appearance and lighter product.

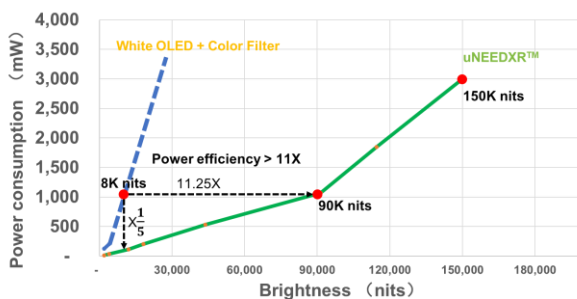


Figure 6. Power consumption comparison for 0.39" XGA

The lifetime of OLEDs is influenced by multiple factors, among which current density is inversely correlated with device

longevity [10]. In our uNEEDXR™ technology and white OLED with color filter approach, our current density is more than five times lower at the same luminance level, as shown in Figure 6. Therefore, our uNEEDXR™ technology can deliver significantly improved lifetime performance.

In terms of wide color gamut, we use the DCI-P3 color space as the standard. Commercial silicon-based OLEDs with white light and color filters generally cover about 81% of the DCI-P3 color gamut, whereas native RGB silicon-based OLEDs can achieve full nearly 100% coverage of the DCI-P3 gamut, as shown in Figure 7.

uNEEDXR™	CIE x	CIE y
R	0.68	0.32
G	0.31	0.67
B	0.14	0.06
DCI-P3	95%	
W-OLED+CF	CIE x	CIE y
R	0.6411	0.3312
G	0.2694	0.6308
B	0.1443	0.0631
DCI-P3	81%	

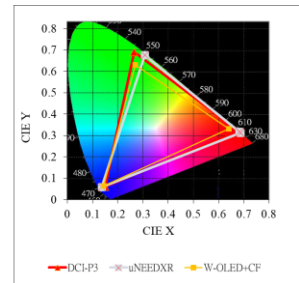


Figure 7. Color gamut (DCI-P3=95%)

4. Conclusion

The system achieves 150,000 nits of high brightness on a 0.39-inch XGA micro-OLED, a high pixel density exceeding 3,000 PPI, an 80% reduction in power consumption compared to commercially available 3,000 nits micro-OLED, an ultra-high contrast ratio of >100,000:1, a 95% DCI-P3 wide color gamut, and precise control over wavelength, bandwidth, and energy distribution (FWHM < 36 nm). Additionally, the system meets customer requirements for reliability and lifespan. This technology provides a scalable production solution for near-eye display applications in augmented reality.

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